The orbit-incidence deficit between separated bands

Human Compression project: L044 continuation

Lemma 1 (Two generating subgroups)**.** *Let $G = \langle P, R\rangle$ act transitively on a finite set Ω of size $n > 1$. Suppose P, R have nonempty supports $Y = \mathrm{supp}(P)$ and $X = \mathrm{supp}(R)$, and respectively p, q nontrivial orbits. Let $H \le P \cap R$, put $W = X \cap Y$, and let $o_W(H)$ count all H-orbits in W, including singleton orbits. Then*

$$X \cup Y = \Omega, \qquad o_W(H) \ge p + q - 1. \tag{1}$$

In particular every nontrivial P- or R-orbit meets W.

Proof. Form the bipartite multigraph whose vertices are all P-orbits and all R-orbits, with one edge for each point of Ω joining the two orbits that contain it. It is connected: a path through successive orbit nodes is exactly a chain of moves by P and R, and these generate the transitive group G. A point fixed by both subgroups would be fixed by G, so no such point exists and $X \cup Y = \Omega$.

A singleton orbit node is a leaf. Delete all such nodes and their incident edges. The remaining nodes are the $p+q$ nontrivial orbits, and the remaining edges are precisely the points of W. This graph is connected: a path between two retained nodes cannot have a leaf as an internal node. Both sorts of retained nodes exist by the nonempty-support assumption. Thus no retained node is isolated. Since $H \le P \cap R$, each of its orbits in W is contained in one intersection of a P-orbit and an R-orbit. Replace the parallel edges belonging to each such H-orbit by one edge. This preserves connectivity, and gives a connected multigraph with $p + q$ vertices and $o_W(H)$ edges. The elementary spanning-tree bound proves (1). □

Theorem 2 (Separated interval with outside budgets)**.** *Let $G = \langle \rho_0, \dots, \rho_{r-1}\rangle$ be a transitive string involution group whose maximal omissions are intransitive. Let $0 \le h < g \le r - 1$, and put*

$$P = G_{<g}, \quad R = G_{>h}, \quad H = G_{\{h+1,\dots,g-1\}}, \quad Y = \mathrm{supp}(P), \quad X = \mathrm{supp}(R).$$

The empty interval means $H = 1$. Suppose each colour strictly between h and g has two ambient-eligible edges, and

$$2h \le n - |X| - 1, \qquad 2(r - 1 - g) \le n - |Y| - 1. \tag{2}$$

If p, q are the numbers of nontrivial P, R-orbits, then

$$2r \le n + 3 - p - q. \tag{3}$$

Consequently, if both groups are intransitive on their supports, then $r \le \lfloor (n-1)/2 \rfloor$.

Proof. The index sets of P, R cover every generator because $g > h$, so $G = \langle P, R\rangle$. Both supports are nonempty. Moreover $H \le P \cap R$; no intersection property is needed for this inclusion. Set $W = X \cap Y$ and $V = \mathrm{supp}(H)$. Then $V \subseteq W$.

For a nonempty middle interval, H lies in a maximal omission and is therefore intransitive on Ω. The complete accepted ambient interval theorem gives, with c its number of nontrivial orbits,

$$2(g - h - 1) \le |V| - c = |W| - o_W(H) \le |W| - p - q + 1. \tag{4}$$

The equality includes every H-fixed point in W; such points are not silently discarded. The last inequality is Lemma 1. If $g = h + 1$, then $H = 1$, $V = \varnothing$ and $|W| - o_W(H) = 0$, so (4) remains true with left side zero. Adding (2) and (4), and using $|X| + |Y| = n + |W|$, gives

$$\begin{aligned} 2r &= 2h + 2(g - h - 1) + 2(r - 1 - g) + 4 \\ &\leq (n - |X| - 1) + (|W| - p - q + 1) + (n - |Y| - 1) + 4 \\ &= n + 3 - p - q. \end{aligned}$$

When both groups are intransitive on their supports, $p, q \geq 2$, so $2r \leq n - 1$ as required. □

Corollary 3 (Oriented two-bridge conclusion)**.** *Use the standing $G \leq A_n$, $n \geq 12$, full string-IP hypotheses of the accepted nested-bridge theorem, including its nontrivial outside regions and prefix/suffix orientation. Suppose h is last shared across the first cut and g first shared across the second, with $i < h < j$, $g < j$, and the stated pure outside actions. Suppose also that every colour strictly between h and g when $g > h$ has two ambient-eligible edges, and the accepted outside-support bounds apply in both directions. Under the same smaller-degree alternating induction as the accepted transitive-tail alternative, either*

$$r \leq \lfloor (n - 1)/2 \rfloor,$$

or an actual first or last three-generator subgroup is one of the two Table 2 six-point A_5 markings, in the appropriate orientation.

Proof. Assume the rank bound fails. The accepted nested-bridge count gives $g \geq h$, and its accepted interior and endpoint touching exclusions give $g > h$. The assumed support bounds are exactly (2), with the reversed prefix index $r - 1 - g$. If both P, R are intransitive on their supports, Theorem 2 contradicts rank failure. Otherwise apply the complete accepted transitive-tail alternative to the first cut if R is transitive, or in the literal reversed orientation to the second cut if P is transitive. Its exact unique-crossing, shared-colour and orientation hypotheses are included here. It supplies precisely the stated actual end subgroup. □

Attribution and remaining source routing. This replaces the last orbit count in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.28, physical page 37. The source uses the support overlap and intransitivity strategy. The explicit orbit-incidence multigraph and the $p + q - 1$ deficit are the present replacement argument. They retain singleton intersection orbits and the corrected reversed index, and avoid assuming that two intransitive tails yield only a two-orbit middle deficit.

All ambient-interval, nested-bridge, touching and transitive-tail proofs and their complete required programs remain paid. No new finite computation or formal declaration is needed. The corollary is deliberately conditioned on the actual cut orientations and support bounds: their complete routing from the source's maximal choice, and its more specific choice of end, remain separate. This packet does not claim whole L044 recovery, new original, question or priority credit.